

Problem 1

- (a) • ($\implies$) Assume f is continuous. Let $A \subseteq X$

Let $x \in \bar{A}$ thus $f(x) \in f(\bar{A})$. Let U be a neighborhood of $f(x)$. Since f is continuous we know that $f^{-1}(U)$ is open and thus a neighborhood of x . From class since $x \in \bar{A} \implies f^{-1}(U) \cap A \neq \emptyset$. Let $y \in f^{-1}(U) \cap A$ thus, $f(y) \in U \cap f(A)$. Thus, U intersects $f(A)$. Therefore, since every neighborhood of $f(x)$ intersects $f(A)$, $f(x) \in \overline{f(A)}$.

- ($\impliedby$) Assume for $A \subseteq X$ that $f(\bar{A}) \subseteq \overline{f(A)}$

Let C be a closed set in Y . Let $A = f^{-1}(C)$. Thus we know that $f(\bar{A}) \subseteq \overline{f(A)}$. Since $f(A) \subseteq C$. Let $x \in \overline{f(A)}$ thus for a neighborhood U of x we know that $U \cap f(A) \neq \emptyset$ and since $f(A) \subseteq C$ we know that $U \cap C \neq \emptyset$. Thus, $x \in \bar{C} = C$. Thus, $\overline{f(A)} \subseteq C$. Therefore, $f(\bar{A}) \subseteq C$. Let $x \in \bar{A}$. Then $f(x) \in C$. Thus, $x \in f^{-1}(C) = A$. Thus, $x \in \bar{A}$. Thus, $\bar{A} \subseteq A \implies \bar{A} = A$. Therefore, A is closed and thus f is continuous.

- (b) Assume X is metrizable with the metric d that induces its topology

- ($\implies$) Assume f is continuous. Let x_n be a sequence in X that converges to x . Let U be a neighborhood of $f(x)$. Thus, $f^{-1}(U)$ is open neighborhood of x . Thus there exists $\epsilon > 0$ s.t. $B_d(x, \epsilon) \subseteq f^{-1}(U)$, and since $x_n \rightarrow x$ we know that $\exists N \in \mathbb{N}$ s.t. $x_n \in B_d(x, \epsilon)$ for all $n \geq N$. Thus consider $f(B_d(x, \epsilon)) \subseteq U$. Thus the same N works s.t. $f(x_n) \in U$ for all $n \geq N$. Thus $f(x_n) \rightarrow f(x)$.
- ($\impliedby$) Assume f is not continuous this means that f is not continuous at some point $x \in X$. Thus, there exists an open neighborhood V of $f(x)$ s.t. for every $\epsilon > 0$ there is a point $y \in B_d(x, \epsilon)$ but $f(y) \notin V$. Construct the sequence s.t. $x_n \in B_d(x, 1/n)$ and $f(x_n) \notin V$. Let U be an open neighborhood of x thus there is a $\epsilon > 0$ s.t. $B_d(x, \epsilon) \subseteq U$. Fix $N \in \mathbb{N}$ s.t. $1/N < \epsilon$ thus for all $n \geq N$ we have that $x_n \in B_d(x, 1/n) \subseteq B_d(x, 1/N) \subseteq B_d(x, \epsilon) \subseteq U$. Thus $x_n \rightarrow x$. However $f(x_n)$ does not converge to $f(x)$ since for all $n \in \mathbb{N}$ we have that $f(x_n) \notin V$.

Problem 2

- (a)
- $\emptyset \in \mathcal{T}$ and $\mathbb{R} \setminus \mathbb{R} = \emptyset$ which is countable thus $\mathbb{R} \in \mathcal{T}$
 - Let $\{U_\alpha\}_{\alpha \in J}$ be a collection of open sets. Consider:

$$U = \bigcup_{\alpha \in J} U_\alpha$$

Thus,

$$\mathbb{R} \setminus \bigcup_{\alpha \in J} U_\alpha = \bigcap_{\alpha \in J} (\mathbb{R} \setminus U_\alpha)$$

Since $\mathbb{R} \setminus U_\alpha$ is countable we know that the arbitrary intersection of countable sets is countable thus we know that U is open.

- Let U, V be open sets consider $U \cap V$. Thus $\mathbb{R} \setminus U \cap V = (\mathbb{R} \setminus U) \cup (\mathbb{R} \setminus V)$, and since finite unions of countable sets is countable we know that $U \cap V$ is open.
- (b) Let $A = \mathbb{R}$ and $0 \in \bar{\mathbb{R}} = \mathbb{R}$. Assume there is a sequence $(x_n) \in \mathbb{R}$ s.t. $x_n \rightarrow 0$. Let $S = \{x_n \mid n \in \mathbb{N}\}$. Consider $U = \mathbb{R} \setminus S$. Since S is countable U is open.

If $0 \notin S$ then $0 \in U$ then U is an open neighborhood of 0. However, for all $n \in \mathbb{N}$ $x_n \notin U$ thus x_n does not converge to 0.

If $0 \notin U$ then $0 \in S$ consider the open set $U' = \mathbb{R} \setminus (S \setminus \{0\})$ since $S \setminus \{0\}$ is countable. Thus use the same logic as the last part since $x_n \notin U$ for all n we know that x_n does not converge to 0.

Problem 3

Assume A is a subspace of a Hausdorff space X . Let $u, v \in A \subseteq X$ thus since X is Hausdorff we know that there are neighborhoods U of u and V of v s.t. $U \cap V = \emptyset$. Thus consider the open neighborhoods $U \cap A$ of u and $V \cap A$ of v in A . Thus consider $(U \cap A) \cap (V \cap A) = (U \cap V) \cap A = \emptyset$. Thus, the subspace A is Hausdorff.

Consider $\mathbb{R}$ with standard topology and consider the equivalence relation $x \sim y$ s.t. $x, y \in [0, \infty)$ or $x, y \in (-\infty, 0)$.

- $x \sim x$ since x will be in the same set as itself
- If $x \sim y$ then $y \sim x$ since x, y are in the same set, so the definition works for both.
- If $x \sim y$ and $y \sim z$ then x, y are in the same set, and y, z are in the same set thus x, z are in the same set.

Thus $\mathbb{R}/\sim = \{A, B\}$ s.t. $\varphi_{\sim}^{-1}(\{A\}) = A = (-\infty, 0)$ and $\varphi_{\sim}^{-1}(\{B\}) = B = [0, \infty)$. Thus we can see that the only open sets are $\{A\}, \{A, B\}, \emptyset$. Thus take points A, B the only open neighborhood of B is $\{A, B\}$ and the open neighborhoods of A are $\{A\}, \{A, B\}$ thus there are no neighborhoods s.t. $U \cap V = \emptyset$. Thus this space is not Hausdorff.

Problem 4

- (a) • ($\implies$) Assume X is Hausdorff.

Let $z \in X \times X \setminus \Delta$ thus $z = (x, y)$ for $x \neq y$. Since X is Hausdorff there are open sets U, V with $x \in U$ and $y \in V$ s.t. $U \cap V = \emptyset$. Thus, $(x, y) \in U \times V$ which is a basis set in $X \times X$. Let $(u, v) \in U \times V$ thus $u \neq v$ since $U \cap V = \emptyset$. Thus $U \times V \subseteq X \times X \setminus \Delta$. Thus, $X \times X \setminus \Delta$ since every basis element is contained in a basis element that is a subset of $X \times X \setminus \Delta$. Therefore, Δ is closed.

- ($\impliedby$) Assume Δ is closed; therefore, $X \times X \setminus \Delta$ is open. Let $x, y \in X$ s.t. $x \neq y$. Consider $(x, y) \in X \times X \setminus \Delta$ since this set is open there is a basis element $U \times V$ with U, V be open in X s.t. $(x, y) \in U \times V \subseteq X \times X \setminus \Delta$. Therefore, $U \cap V = \emptyset$, since if it wasn't empty then $(U \times V) \cap \Delta \neq \emptyset$. Thus, $x \in U$ and $y \in V$ s.t. $U \cap V = \emptyset$. Thus, X is Hausdorff.

- (b) Assume f, g are continuous maps $X \rightarrow Y$ s.t. $f(x) = g(x)$ for all $x \in U$, and that U is dense. Consider the map $h : X \rightarrow Y \times Y$ s.t. $h(x) = (f(x), g(x))$. Since f, g are continuous we know that h is continuous. From part (a) since Y is Hausdorff we know that $\Delta \subset Y \times Y$ is closed. Thus, we know that $h^{-1}(\Delta)$ is also closed. Since $h^{-1}(\Delta) = \{x \in X \mid (f(x), g(x)) \in \Delta\} = \{x \in X \mid f(x) = g(x)\}$. Thus, $U \subseteq h^{-1}(\Delta)$. However, $h^{-1}(\Delta)$ is a closed set that contains U we know that $\bar{U} = X \subseteq h^{-1}(\Delta)$. Therefore, $h^{-1}(\Delta) = X$. Thus $f = g$

Problem 5

Define $f' : Z \rightarrow Y$ s.t. $f'([x]) = f(x)$

Well Defined:

Since p is surjective, if $[x] = [a]$ this means that $p(x) = p(a)$ thus this means that $f(x) = f(a)$ thus $f'([x]) = f(x) = f(a) = f'([a])$

Consider $f' \circ p$. Let $x \in X$. Thus, $f' \circ p(x) = f'([x]) = f(x)$. Thus $f' \circ p = f$.

Uniqueness:

Assume there is another map g s.t. $g \circ p = f$. Let $[x] \in Z$. Thus, $p(x) = [x]$

$$g([x]) = g(p(x)) = f(x)$$

Also,

$$f'([x]) = f'(p(x)) = f(x)$$

Thus $g = f'$

Assume f' is continuous. Since p is continuous and $f = f' \circ p$ we know that f is continuous.

Assume f is continuous. Let V be open in Y . Consider $(f')^{-1}(V) = U$. For this to be open $p^{-1}(U)$ has to be open. However, since $p^{-1}(U) = \{x \in X \mid f'(p(x)) \in V\} = \{x \in X \mid f(x) \in V\} = f^{-1}(V)$. Since f is continuous we know that $f^{-1}(V)$ is open thus $p^{-1}(U)$ is open thus U is open. Therefore, f' is continuous.

Problem 6

Let p be the point $[(x, y, 0)]$ in X .

Let $y, y' = [(0, 0, -1)]$.

Define a function $f_1 : Y_1 \rightarrow X$ s.t.

$$f_1(x, y, z) = \begin{cases} (x, y, (1+z)/2) / \sqrt{x^2 + y^2 + ((1+z)/2)^2} & (x, y, z) \neq (0, 0, -1), \\ p, & (x, y, z) = (0, 0, -1). \end{cases}$$

Define a function $f_2 : Y_2 \rightarrow X$ s.t.

$$f_2(x, y, z) = \begin{cases} (x, y, -(1+z)/2) / \sqrt{x^2 + y^2 + (-(1+z)/2)^2} & (x, y, z) \neq (0, 0, -1), \\ p, & (x, y, z) = (0, 0, -1). \end{cases}$$

- (Well Defined) for $(x, y, z) \in Y_1 \setminus (0, 0, -1)$ we have that $x^2 + y^2 + z^2 = 1$ and $-1 < z \leq 1$. Thus $(1+z)/2 > 0$ so then this point $(x, y, (1+z)/2)$ is on the upper hemisphere once you normalize it.

Then the south pole is well defined.

- (Injective) Let a, b be points in Y_1 not the south pole s.t. $F(a) = F(b)$ then these only differ by a normalization factor thus $(a_1, a_2, (1+a_3)/2) = k(b_1, b_2, (1+b_3)/2)$ for $k \in \mathbb{R}$. However since they both have the same magnitude before the function this implies that $a = b$

Also since no other point other than $(0, 0, -1)$ goes to p it does not conflict.

Thus Injective

- (Continuous for $Y_1 \setminus (0, 0, -1)$) Since this is just a division of polynomials where the divisor is not zero since the magnitude is 1 we get that f_1 is continuous.
- (Continuous at $(0, 0, -1)$) Let p be the quotient map for X . Let U be an open neighborhood of p thus $p^{-1}(U)$ is open in S^2 which contains the entire equator. Fix $\epsilon > 0$ s.t. $B(s, \epsilon) \cap S^2 \subseteq q^{-1}(U)$ where s is on the equator. Now do this for all the things on the equator and build a new set s.t. $S^2 \cap \bigcup_{s \in E} B(s, \epsilon_s) \subseteq q^{-1}(U)$. Where E is the equator. Fix delta s.t. $\delta < \min(1, \sqrt{10}\epsilon)$ where epsilon is the min of all the balls made up in that union.

Thus, if $\sqrt{x^2 + y^2 + (z+1)^2} = r < \delta$. Thus, $|z+1| = z+1 < \delta \leq 1$ since $-1 \leq z < 1$. Let $h = z+1$ thus $z = -1+h$ Since $x^2 + y^2 + z^2 = 1$ thus $x^2 + y^2 = 2h - h^2$. Thus,

$r^2 = 2h - h^2 + h^2$. Thus, $r^2 = 2h$. Thus, $h = r^2/2$ since $h < 1$ we know that

$$1 - 3h/8 \geq 5/8$$

Thus,

$$2h - 3h^2/4 \geq 10h/8 = 5h/4$$

Thus,

$$\sqrt{2h - 3h^2/4} \geq \sqrt{5h}/2$$

Thus $|(f_1(x, y, z))_3| = (h/2)/\sqrt{2h - 3h^2/4} \leq (h/2)/(\sqrt{5h}/2) = h/\sqrt{5h} = \sqrt{h}/\sqrt{5} \leq r/\sqrt{10} \leq \delta/\sqrt{10} < \epsilon$

Thus, the z coordinate is less than epsilon away from the 0. Thus, $f_1(x, y, z) \in \bigcup_{s \in E} B(s, \epsilon_e) \subseteq q^{-1}(U)$. This makes sense because we are only looking at when f_1 maps to the non-equivalence relation part of X . Thus, since q surjective and $f_1(x, y, z)$ is away from the south pole thus $q(f_1(x, y, z)) = [f_1(x, y, z)] = \{f_1(x, y, z)\}$. Thus, we get $q(f_1(x, y, z)) = f_1(x, y, z) \in q(\bigcup_{s \in E} B(s, \epsilon_e)) \subset U$. Thus let V be the open neighborhood in Y_1 s.t. $V = \{(x, y, z) \in Y_1 \mid |z + 1| < \delta\}$. By above we know this means that the third coordinate of $|f_1(x, y, z)| < \epsilon$ which implies $f_1(x, y, z) \in \bigcup_{s \in E} B(s, \epsilon_e) \subseteq q^{-1}(U)$. Thus, $q(f_1(x, y, z)) = f_1(x, y, z) \in q(\bigcup_{s \in E} B(s, \epsilon_e)) \subseteq U$ since q is surjective. Thus, $f_1(V) \subseteq U$. Thus, we are continuous at $(0, 0, -1)$.

Thus, f_1 is continuous.

- (Embedding) Look at $f_1^{-1} : f_1(Y_1) \rightarrow Y_1$ s.t.

$$f_1^{-1}(u, v, w) \begin{cases} (4wu/(1 + 3w^2), 4wv/(1 + 3w^2), (5w^2 - 1)/(1 + 3w^2)) & \text{if } w \neq 0, \\ (0, 0, -1) & \text{if } w = 0. \end{cases}$$

$$f_1^{-1} \circ f_1(x) = x \text{ and } f \circ f_1^{-1}(x) = x$$

Thus, f^{-1} is continuous on $f_1(Y_1) \setminus [(1, 0, 0)]$ since it is polynomials and rational functions where the denominator is not zero.

Since X is a subspace of $\mathbb{R}^3$ it is metrizable therefore we can check continuous with sequences. Thus consider a sequence $(u_n, v_n, w_n) \rightarrow (u_0, v_0, 0)$. Consider $f_1^{-1}(u_n, v_n, w_n)$. As $w_n \rightarrow 0$ we get that $4w_n u_n / (1 + 3w_n^2) \rightarrow 0$ and same for the second coordinate. For the third we get that $(5w_n^2 - 1)/(1 + 3w_n^2) \rightarrow -1/1 = -1$ as $w_n \rightarrow 0$. Thus f_1^{-1} is continuous at all points on the equator.

Thus, f_1 is an embedding on to the upper hemisphere of X .

Ok do the same thing for f_2

Define $f : Y \rightarrow X$ s.t.

$$f(x) = \begin{cases} f_1(x) & \text{if } x \in Y_1 \\ f_2(x) & \text{if } x \in Y_2 \end{cases}$$

Y_1 is closed in Y since it is the complement of $Y_2 \setminus \{0,0,-1\}$. Then when we bring it back to $Y_1 \sqcup Y_2$ we get that $q^{-1}(Y_2 \setminus \{0,0,-1\}) = \emptyset \sqcup Y_2 \setminus \{0,0,-1\}$. This is open since $Y_2 \setminus \{0,0,-1\}$ is open in Y_2 , since it is equal to $A = \{(x,y,z) \in \mathbb{R}^3 \mid z > -1\} \cap Y_2$ since A is open in $\mathbb{R}^3$ we get that $A \cap Y_2$ is open in Y_2 thus we bring it back and see that $\emptyset \sqcup (A \cap Y_2) = q^{-1}(Y_2 \setminus \{0,0,-1\})$ is open thus $Y_2 \setminus \{0,0,-1\}$ is open thus Y_1 is closed.

Do the same for Y_2 thus,

Using the pasting lemma since f_1 is continuous and f_2 is continuous and f_1, f_2 are continuous on $Y_1 \cap Y_2$. f is continuous.

Injective:

Let $a, b \in Y$ s.t. $f(x) = f(y)$ this either means that $f_1(x) = f_2(y)$ or $f_i(x) = f_i(y)$ where $i \in \{1,2\}$. If $f_1(x) = f_2(y)$ then this is only true when $x = (0,0,-1) = y$. Thus done. If $f_i(x) = f_i(y)$. Then this means $x, y \in Y_i$. Since f_i is injective on Y_i . $x = y$.

Surjective:

Let $x \in X$. If x is in the either the upper hemisphere or lower hemisphere and not the equator. This means that there is a $y \in Y_i$ s.t. $f_i(y) = x$. This is because f_i is bijective from $Y_i \setminus \{(0,0,-1)\} \rightarrow H$ where H is the open hemisphere, and since x is not on the equator we know that $y \neq (0,0,-1)$.

If x is on the equator then pick $f_1(0,0,-1) = x$.

Thus f is bijective.

Let U be open in Y consider $(f^{-1})^{-1}(U) = f(U)$ since f is bijective, since f agrees with f_1 on Y_1 and f_2 on Y_2 we get that $f(U) = f((U \cap Y_1) \cup (U \cap Y_2)) = f_1(U \cap Y_1) \cup f_2(U \cap Y_2)$. Since f_1, f_2 are embeddings of Y_1, Y_2 onto the hemispheres we get that $f_1(U \cap Y_1)$ is open and $f_2(U \cap Y_2)$ is open since $U \cap Y_1$, and $U \cap Y_2$ are open in their topologies. Thus, $f_1(U \cap Y_1) \cup f_2(U \cap Y_2)$ is open thus $f(U)$ is open. Thus, f is a homeomorphism.

Problem 7

Let $x, y \in Y$ s.t. $x \neq y$.

If x, y are in the same sphere lets say Y_1 then since Y_1 is a sub space of $\mathbb{R}^3$ and $\mathbb{R}^3$ is metrizable thus Hausdorff we know that there are sets $x \in U \subseteq \mathbb{R}^3$ and $y \in V \subseteq \mathbb{R}^3$ that are open s.t. $U \cap V = \emptyset$. Thus $(U \cap Y_1) \cap (V \cap Y_1) = \emptyset$. Thus, $(U \cap Y_1) \sqcup \emptyset$ and $\emptyset \sqcup (U \cap Y_2)$. Thus, $q(U \cap Y_1) \cap q(V \cap Y_1) = \emptyset$, and these are open sets with $[x], [y]$ in them since q is surjective and $q^{-1}(q(H)) = H$

If x, y are in different spheres and are not the gluing point. Let $x \in U$ be open in Y_1 and $y \in V$ be open in Y_2 and since the spheres are subspaces of a Hausdorff space we do similar to case 1, and we can pick the open sets s.t. $U \cap K$ and open neighborhood of $\{0, 0, -1\}$ is empty. Do the same for V thus now we know that $U \sqcup \emptyset$ is open in $Y_1 \sqcup Y_2$ and same for $\emptyset \sqcup V$. Thus the intersection of these is empty then we apply q and do the same thing as the last case.

If one of x, y is the gluing point. Then they are in the same sphere, and since it is a subspace of a Hausdorff space we do the same thing as case 1 and pick $x \in U$ open in Y_1 and $y \in V$ open in Y_1 s.t. $U \cap V$ is empty. Thus, now if y is the gluing point then consider the set $U \sqcup \emptyset$ and $V \sqcup Y_2$ thus these are open in $Y_1 \sqcup Y_2$ with an empty intersection then apply q to them and we get that they are open with empty intersection with $[y] \in q(V \sqcup Y_2)$ and $[x] \in q(U \sqcup \emptyset)$